

```

!pip install apyori

import pandas as pd
from apyori import apriori

data = pd.read_csv("/content/Market_Basket_Optimisation.csv",
header=None)

transactions = []
for i in range(0, len(data)):
    transaction = []
    for j in range(0, 20):
        item = str(data.values[i, j])
        if item != 'nan':
            transaction.append(item)
    transactions.append(transaction)

rules = apriori(transactions,
                 min_support=0.003,
                 min_confidence=0.2,
                 min_lift=3,
                 min_length=2)

results = list(rules)

for rule in results:
    items = list(rule.items)
    support = rule.support

    for stat in rule.ordered_statistics:
        antecedent = list(stat.items_base)
        consequent = list(stat.items_add)
        confidence = stat.confidence
        lift = stat.lift

        if len(antecedent) > 0 and len(consequent) > 0:
            print("Rule:", antecedent, "→", consequent)
            print("Support:", round(support, 4))
            print("Confidence:", round(confidence, 4))
            print("Lift:", round(lift, 4))
            print("-" * 40)

```

Collecting apyori

Downloading apyori-1.1.2.tar.gz (8.6 kB)

Preparing metadata (setup.py) ... e=apyori-1.1.2-py3-none-any.whl
size=5954

sha256=c793d3c96bb324915d4574741991658f9adadec1bfff5a53a527afabe943e510
0

Stored in directory:

/root/.cache/pip/wheels/7f/49/e3/42c73b19a264de37129fadaa0c52f26cf50e87de08fb9804af

Successfully built apyori

Installing collected packages: apyori

Successfully installed apyori-1.1.2

Rule: ['light cream'] → ['chicken']

Support: 0.0045

Confidence: 0.2906

Lift: 4.844

Rule: ['mushroom cream sauce'] → ['escalope']

Support: 0.0057

Confidence: 0.3007

Lift: 3.7908

Rule: ['pasta'] → ['escalope']

Support: 0.0059

Confidence: 0.3729

Lift: 4.7008

Rule: ['fromage blanc'] → ['honey']

Support: 0.0033

Confidence: 0.2451

Lift: 5.1643

Rule: ['herb & pepper'] → ['ground beef']

Support: 0.016

Confidence: 0.3235

Lift: 3.292

Rule: ['tomato sauce'] → ['ground beef']

Support: 0.0053

Confidence: 0.3774

Lift: 3.8407

Rule: ['light cream'] → ['olive oil']

Support: 0.0032

Confidence: 0.2051

Lift: 3.1147

Rule: ['whole wheat pasta'] → ['olive oil']

Support: 0.008

Confidence: 0.2715

Lift: 4.1224

Rule: ['pasta'] → ['shrimp']

Support: 0.0051

Confidence: 0.322

Lift: 4.5067

Rule: ['spaghetti', 'avocado'] → ['milk']
Support: 0.0033
Confidence: 0.4167
Lift: 3.2154

Rule: ['cake', 'milk'] → ['burgers']
Support: 0.0037
Confidence: 0.28
Lift: 3.2114

Rule: ['turkey', 'chocolate'] → ['burgers']
Support: 0.0031
Confidence: 0.2706
Lift: 3.1035

Rule: ['turkey', 'milk'] → ['burgers']
Support: 0.0032
Confidence: 0.2824
Lift: 3.2384

Rule: ['frozen vegetables', 'cake'] → ['tomatoes']
Support: 0.0031
Confidence: 0.2987
Lift: 4.3676

Rule: ['cake', 'tomatoes'] → ['frozen vegetables']
Support: 0.0031
Confidence: 0.3651
Lift: 3.83

Rule: ['ground beef', 'cereals'] → ['spaghetti']
Support: 0.0031
Confidence: 0.6765
Lift: 3.8853

Rule: ['spaghetti', 'cereals'] → ['ground beef']
Support: 0.0031
Confidence: 0.46
Lift: 4.6818

Rule: ['chicken', 'ground beef'] → ['milk']
Support: 0.0039
Confidence: 0.4085
Lift: 3.152

Rule: ['chicken', 'milk'] → ['olive oil']
Support: 0.0036
Confidence: 0.2432

Lift: 3.6935

Rule: ['chicken', 'olive oil'] → ['milk']

Support: 0.0036

Confidence: 0.5

Lift: 3.8585

Rule: ['olive oil', 'milk'] → ['chicken']

Support: 0.0036

Confidence: 0.2109

Lift: 3.5161

Rule: ['chicken', 'spaghetti'] → ['olive oil']

Support: 0.0035

Confidence: 0.2016

Lift: 3.0604

Rule: ['frozen vegetables', 'chocolate'] → ['shrimp']

Support: 0.0053

Confidence: 0.2326

Lift: 3.2545

Rule: ['shrimp', 'chocolate'] → ['frozen vegetables']

Support: 0.0053

Confidence: 0.2963

Lift: 3.1084

Rule: ['herb & pepper', 'chocolate'] → ['ground beef']

Support: 0.004

Confidence: 0.4412

Lift: 4.4902

Rule: ['soup', 'chocolate'] → ['milk']

Support: 0.004

Confidence: 0.3947

Lift: 3.0462

Rule: ['ground beef', 'cooking oil'] → ['spaghetti']

Support: 0.0048

Confidence: 0.5714

Lift: 3.282

Rule: ['spaghetti', 'cooking oil'] → ['ground beef']

Support: 0.0048

Confidence: 0.3025

Lift: 3.079

Rule: ['ground beef', 'eggs'] → ['herb & pepper']

Support: 0.0041

Confidence: 0.2067

Lift: 4.1785

Rule: ['herb & pepper', 'eggs'] → ['ground beef']

Support: 0.0041

Confidence: 0.3298

Lift: 3.3565

Rule: ['red wine', 'eggs'] → ['spaghetti']

Support: 0.0037

Confidence: 0.5283

Lift: 3.0343

Rule: ['french fries', 'ground beef'] → ['herb & pepper']

Support: 0.0032

Confidence: 0.2308

Lift: 4.6658

Rule: ['french fries', 'herb & pepper'] → ['ground beef']

Support: 0.0032

Confidence: 0.4615

Lift: 4.6974

Rule: ['frozen vegetables', 'green tea'] → ['tomatoes']

Support: 0.0033

Confidence: 0.2315

Lift: 3.3847

Rule: ['frozen vegetables', 'spaghetti'] → ['ground beef']

Support: 0.0087

Confidence: 0.311

Lift: 3.1653

Rule: ['frozen vegetables', 'milk'] → ['olive oil']

Support: 0.0048

Confidence: 0.2034

Lift: 3.0883

Rule: ['frozen vegetables', 'olive oil'] → ['milk']

Support: 0.0048

Confidence: 0.4235

Lift: 3.2684

Rule: ['soup', 'frozen vegetables'] → ['milk']

Support: 0.004

Confidence: 0.5

Lift: 3.8585

Rule: ['milk', 'tomatoes'] → ['frozen vegetables']

Support: 0.0041

Confidence: 0.2952

Lift: 3.0973

Rule: ['mineral water', 'shrimp'] → ['frozen vegetables']

Support: 0.0072

Confidence: 0.3051

Lift: 3.2006

Rule: ['frozen vegetables', 'spaghetti'] → ['olive oil']

Support: 0.0057

Confidence: 0.2057

Lift: 3.124

Rule: ['frozen vegetables', 'spaghetti'] → ['shrimp']

Support: 0.006

Confidence: 0.2153

Lift: 3.0131

Rule: ['frozen vegetables', 'shrimp'] → ['tomatoes']

Support: 0.004

Confidence: 0.24

Lift: 3.5092

Rule: ['frozen vegetables', 'tomatoes'] → ['shrimp']

Support: 0.004

Confidence: 0.2479

Lift: 3.4697

Rule: ['shrimp', 'tomatoes'] → ['frozen vegetables']

Support: 0.004

Confidence: 0.3571

Lift: 3.7468

Rule: ['frozen vegetables', 'spaghetti'] → ['tomatoes']

Support: 0.0067

Confidence: 0.2392

Lift: 3.498

Rule: ['spaghetti', 'tomatoes'] → ['frozen vegetables']

Support: 0.0067

Confidence: 0.3185

Lift: 3.3411

Rule: ['grated cheese', 'spaghetti'] → ['ground beef']

Support: 0.0053

Confidence: 0.3226

Lift: 3.2831

Rule: ['green tea', 'ground beef'] → ['tomatoes']
Support: 0.0031
Confidence: 0.2072
Lift: 3.0297

Rule: ['herb & pepper', 'milk'] → ['ground beef']
Support: 0.0036
Confidence: 0.3913
Lift: 3.9826

Rule: ['mineral water', 'herb & pepper'] → ['ground beef']
Support: 0.0067
Confidence: 0.3906
Lift: 3.9757

Rule: ['spaghetti', 'herb & pepper'] → ['ground beef']
Support: 0.0064
Confidence: 0.3934
Lift: 4.0044

Rule: ['ground beef', 'milk'] → ['olive oil']
Support: 0.0049
Confidence: 0.2242
Lift: 3.4049

Rule: ['soup', 'ground beef'] → ['milk']
Support: 0.004
Confidence: 0.411
Lift: 3.1714

Rule: ['pepper', 'spaghetti'] → ['ground beef']
Support: 0.0033
Confidence: 0.3378
Lift: 3.4384

Rule: ['shrimp', 'ground beef'] → ['spaghetti']
Support: 0.006
Confidence: 0.5233
Lift: 3.0053

Rule: ['tomato sauce'] → ['spaghetti', 'ground beef']
Support: 0.0031
Confidence: 0.217
Lift: 5.536

Rule: ['tomato sauce', 'ground beef'] → ['spaghetti']
Support: 0.0031
Confidence: 0.575
Lift: 3.3025

Rule: ['spaghetti', 'tomato sauce'] → ['ground beef']
Support: 0.0031
Confidence: 0.4894
Lift: 4.9806

Rule: ['light cream'] → ['mineral water', 'spaghetti']
Support: 0.0032
Confidence: 0.2051
Lift: 3.4345

Rule: ['shrimp', 'olive oil'] → ['milk']
Support: 0.0032
Confidence: 0.3934
Lift: 3.0362

Rule: ['olive oil', 'milk'] → ['soup']
Support: 0.0036
Confidence: 0.2109
Lift: 4.1748

Rule: ['soup', 'milk'] → ['olive oil']
Support: 0.0036
Confidence: 0.2368
Lift: 3.5963

Rule: ['soup', 'olive oil'] → ['milk']
Support: 0.0036
Confidence: 0.403
Lift: 3.1099

Rule: ['spaghetti', 'milk'] → ['olive oil']
Support: 0.0072
Confidence: 0.203
Lift: 3.0825

Rule: ['milk', 'tomatoes'] → ['soup']
Support: 0.0031
Confidence: 0.219
Lift: 4.3353

Rule: ['soup', 'tomatoes'] → ['milk']
Support: 0.0031
Confidence: 0.4423
Lift: 3.4133

Rule: ['spaghetti', 'whole wheat pasta'] → ['milk']
Support: 0.004
Confidence: 0.4545

Lift: 3.5078

Rule: ['soup', 'mineral water'] → ['olive oil']

Support: 0.0052

Confidence: 0.2254

Lift: 3.423

Rule: ['mineral water', 'whole wheat pasta'] → ['olive oil']

Support: 0.0039

Confidence: 0.4028

Lift: 6.1159

Rule: ['spaghetti', 'pancakes'] → ['olive oil']

Support: 0.0051

Confidence: 0.2011

Lift: 3.0529

Rule: ['olive oil', 'tomatoes'] → ['spaghetti']

Support: 0.0044

Confidence: 0.6111

Lift: 3.5099

Rule: ['spaghetti', 'tomatoes'] → ['olive oil']

Support: 0.0044

Confidence: 0.2102

Lift: 3.1916

Rule: ['spaghetti', 'whole wheat rice'] → ['tomatoes']

Support: 0.0031

Confidence: 0.217

Lift: 3.1727

Rule: ['ground beef', 'eggs'] → ['mineral water', 'chocolate']

Support: 0.004

Confidence: 0.2

Lift: 3.798

Rule: ['mineral water', 'chocolate', 'eggs'] → ['ground beef']

Support: 0.004

Confidence: 0.297

Lift: 3.0231

Rule: ['mineral water', 'frozen vegetables', 'chocolate'] → ['ground beef']

Support: 0.0033

Confidence: 0.3425

Lift: 3.4855

Rule: ['mineral water', 'ground beef', 'chocolate'] → ['frozen

```
vegetables']
Support: 0.0033
Confidence: 0.3049
Lift: 3.1984
-----
Rule: ['frozen vegetables', 'ground beef', 'chocolate'] →
['spaghetti']
Support: 0.0031
Confidence: 0.5349
Lift: 3.0721
-----
Rule: ['frozen vegetables', 'spaghetti', 'chocolate'] → ['ground
beef']
Support: 0.0031
Confidence: 0.3898
Lift: 3.9676
-----
Rule: ['spaghetti', 'ground beef', 'chocolate'] → ['frozen
vegetables']
Support: 0.0031
Confidence: 0.3333
Lift: 3.497
-----
Rule: ['mineral water', 'frozen vegetables', 'chocolate'] → ['milk']
Support: 0.004
Confidence: 0.411
Lift: 3.1714
-----
Rule: ['frozen vegetables', 'spaghetti', 'chocolate'] → ['milk']
Support: 0.0035
Confidence: 0.4407
Lift: 3.4007
-----
Rule: ['spaghetti', 'chocolate', 'milk'] → ['frozen vegetables']
Support: 0.0035
Confidence: 0.3171
Lift: 3.3264
-----
Rule: ['mineral water', 'frozen vegetables', 'chocolate'] → ['shrimp']
Support: 0.0032
Confidence: 0.3288
Lift: 4.6009
-----
Rule: ['mineral water', 'shrimp', 'chocolate'] → ['frozen vegetables']
Support: 0.0032
Confidence: 0.4211
Lift: 4.4172
-----
Rule: ['olive oil', 'chocolate'] → ['mineral water', 'spaghetti']
```

Support: 0.0039

Confidence: 0.2358

Lift: 3.9476

Rule: ['mineral water', 'spaghetti', 'chocolate'] → ['olive oil']

Support: 0.0039

Confidence: 0.2437

Lift: 3.7004

Rule: ['mineral water', 'spaghetti', 'chocolate'] → ['shrimp']

Support: 0.0035

Confidence: 0.2185

Lift: 3.0576

Rule: ['mineral water', 'frozen vegetables', 'eggs'] → ['milk']

Support: 0.0037

Confidence: 0.4118

Lift: 3.1776

Rule: ['milk', 'frozen smoothie'] → ['mineral water', 'spaghetti']

Support: 0.0032

Confidence: 0.2243

Lift: 3.7555

Rule: ['spaghetti', 'frozen smoothie'] → ['mineral water', 'milk']

Support: 0.0032

Confidence: 0.2051

Lift: 4.2741

Rule: ['mineral water', 'spaghetti', 'frozen smoothie'] → ['milk']

Support: 0.0032

Confidence: 0.4706

Lift: 3.6316

Rule: ['mineral water', 'spaghetti', 'milk'] → ['frozen smoothie']

Support: 0.0032

Confidence: 0.2034

Lift: 3.2118

Rule: ['frozen vegetables', 'ground beef'] → ['mineral water', 'milk']

Support: 0.0037

Confidence: 0.2205

Lift: 4.5938

Rule: ['mineral water', 'frozen vegetables', 'ground beef'] → ['milk']

Support: 0.0037

Confidence: 0.4058

Lift: 3.1316

Rule: ['mineral water', 'frozen vegetables', 'milk'] → ['ground beef']

Support: 0.0037

Confidence: 0.3373

Lift: 3.4335

Rule: ['mineral water', 'ground beef', 'milk'] → ['frozen vegetables']

Support: 0.0037

Confidence: 0.3373

Lift: 3.5391

Rule: ['frozen vegetables', 'ground beef', 'milk'] → ['spaghetti']

Support: 0.0031

Confidence: 0.5349

Lift: 3.0721

Rule: ['frozen vegetables', 'spaghetti', 'milk'] → ['ground beef']

Support: 0.0031

Confidence: 0.371

Lift: 3.7756

Rule: ['spaghetti', 'ground beef', 'milk'] → ['frozen vegetables']

Support: 0.0031

Confidence: 0.3151

Lift: 3.3054

Rule: ['frozen vegetables', 'ground beef'] → ['mineral water',
'spaghetti']

Support: 0.0044

Confidence: 0.2598

Lift: 4.3506

Rule: ['mineral water', 'frozen vegetables', 'spaghetti'] → ['ground
beef']

Support: 0.0044

Confidence: 0.3667

Lift: 3.7318

Rule: ['frozen vegetables', 'olive oil'] → ['mineral water', 'milk']

Support: 0.0033

Confidence: 0.2941

Lift: 6.1283

Rule: ['mineral water', 'frozen vegetables', 'milk'] → ['olive oil']

Support: 0.0033

Confidence: 0.3012

Lift: 4.5736

Rule: ['mineral water', 'frozen vegetables', 'olive oil'] → ['milk']

Support: 0.0033

Confidence: 0.5102

Lift: 3.9373

Rule: ['mineral water', 'olive oil', 'milk'] → ['frozen vegetables']

Support: 0.0033

Confidence: 0.3906

Lift: 4.098

Rule: ['soup', 'frozen vegetables'] → ['mineral water', 'milk']

Support: 0.0031

Confidence: 0.3833

Lift: 7.9872

Rule: ['soup', 'milk'] → ['mineral water', 'frozen vegetables']

Support: 0.0031

Confidence: 0.2018

Lift: 5.6469

Rule: ['mineral water', 'frozen vegetables', 'milk'] → ['soup']

Support: 0.0031

Confidence: 0.2771

Lift: 5.4844

Rule: ['soup', 'frozen vegetables', 'milk'] → ['mineral water']

Support: 0.0031

Confidence: 0.7667

Lift: 3.2163

Rule: ['mineral water', 'frozen vegetables', 'soup'] → ['milk']

Support: 0.0031

Confidence: 0.6053

Lift: 4.6709

Rule: ['soup', 'mineral water', 'milk'] → ['frozen vegetables']

Support: 0.0031

Confidence: 0.3594

Lift: 3.7702

Rule: ['mineral water', 'spaghetti', 'milk'] → ['frozen vegetables']

Support: 0.0045

Confidence: 0.2881

Lift: 3.0228

Rule: ['frozen vegetables', 'shrimp'] → ['mineral water', 'spaghetti']

Support: 0.0033

Confidence: 0.2

Lift: 3.3487

Rule: ['mineral water', 'frozen vegetables', 'spaghetti'] → ['shrimp']

Support: 0.0033

Confidence: 0.2778

Lift: 3.8873

Rule: ['spaghetti', 'mineral water', 'shrimp'] → ['frozen vegetables']

Support: 0.0033

Confidence: 0.3906

Lift: 4.098

Rule: ['mineral water', 'frozen vegetables', 'spaghetti'] →
['tomatoes']

Support: 0.0031

Confidence: 0.2556

Lift: 3.7367

Rule: ['mineral water', 'frozen vegetables', 'tomatoes'] →
['spaghetti']

Support: 0.0031

Confidence: 0.5227

Lift: 3.0023

Rule: ['mineral water', 'spaghetti', 'tomatoes'] → ['frozen
vegetables']

Support: 0.0031

Confidence: 0.3286

Lift: 3.447

Rule: ['ground beef', 'milk'] → ['mineral water', 'spaghetti']

Support: 0.0044

Confidence: 0.2

Lift: 3.3487

Rule: ['ground beef', 'olive oil'] → ['mineral water', 'spaghetti']

Support: 0.0031

Confidence: 0.217

Lift: 3.633

Rule: ['mineral water', 'spaghetti', 'olive oil'] → ['ground beef']

Support: 0.0031

Confidence: 0.2987

Lift: 3.0401

Rule: ['ground beef', 'pancakes'] → ['mineral water', 'spaghetti']

Support: 0.0031

Confidence: 0.211

Lift: 3.533

Rule: ['ground beef', 'tomatoes'] → ['mineral water', 'spaghetti']

Support: 0.0031

Confidence: 0.2614

Lift: 4.3761

Rule: ['mineral water', 'ground beef', 'tomatoes'] → ['spaghetti']

Support: 0.0031

Confidence: 0.561

Lift: 3.222

Rule: ['mineral water', 'spaghetti', 'tomatoes'] → ['ground beef']

Support: 0.0031

Confidence: 0.3286

Lift: 3.3441

Rule: ['mineral water', 'spaghetti', 'milk'] → ['olive oil']

Support: 0.0033

Confidence: 0.2119

Lift: 3.217

Rule: ['milk', 'tomatoes'] → ['mineral water', 'spaghetti']

Support: 0.0033

Confidence: 0.2381

Lift: 3.9865

Rule: ['mineral water', 'spaghetti', 'milk'] → ['tomatoes']

Support: 0.0033

Confidence: 0.2119

Lift: 3.0978
